API Inspector - Mini Project

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S6 CSD

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Introduction

- API Unit Tester is a specialized software testing tool that ensures API endpoints function correctly.
- It allows developers to automate the validation of API requests and responses.
- Provides detailed reporting, debugging support, and error simulation for better testing.

Problem Statement

- APIs are widely used in modern applications, connecting various services.
- Manual testing of APIs is inefficient, time-consuming, and prone to human errors.
- Need for an automated API testing framework that simplifies and accelerates the process.

Objectives

- Ensure API reliability through automated validation.
- Reduce manual effort and accelerate the testing cycle.
- Detect potential failures early through error simulation.
- Provide detailed logs and reports for debugging.

Purpose and Need

- APIs form the backbone of modern applications.
- Ensuring their reliability is crucial for smooth communication between services.
- Automated testing minimizes production issues and improves user experience.

Scope

- Supports unit testing for various HTTP methods.
- Enables error handling and invalid data inputs.
- Provides authentication handling for secure API testing.
- Generates comprehensive logs and reports.

Social Relevance

- Enhances software quality and reliability.
- Reduces the number of API-related bugs.
- Saves time and resources by automating tedious testing tasks.

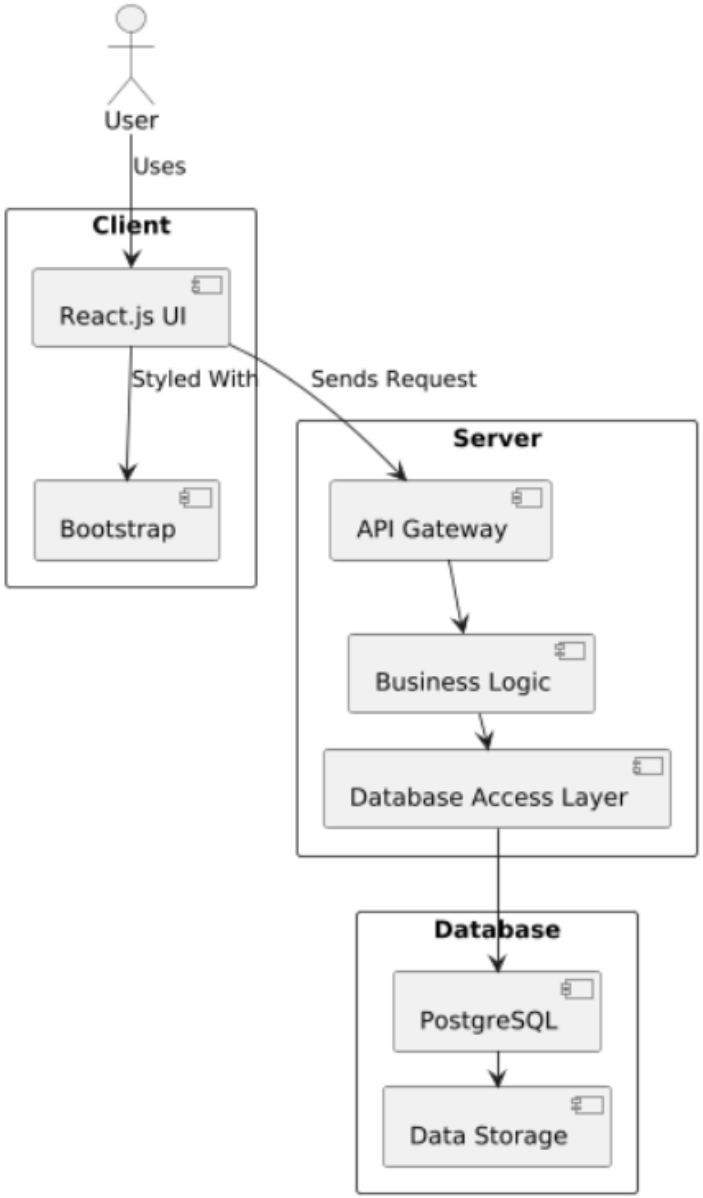
Software Requirement Specification

- Functional Requirements: Request Handling, Authentication, Test Execution, Error Simulation, Reporting.
- Non-functional Requirements: Performance, Scalability, Security, Reliability.

Technology Stack

- Frontend: React.js,Bootstrap.
- Backend: Node.js with Express
- Database: PostgreSQL.
- Testing Frameworks: Mocha, Jest, Supertest.

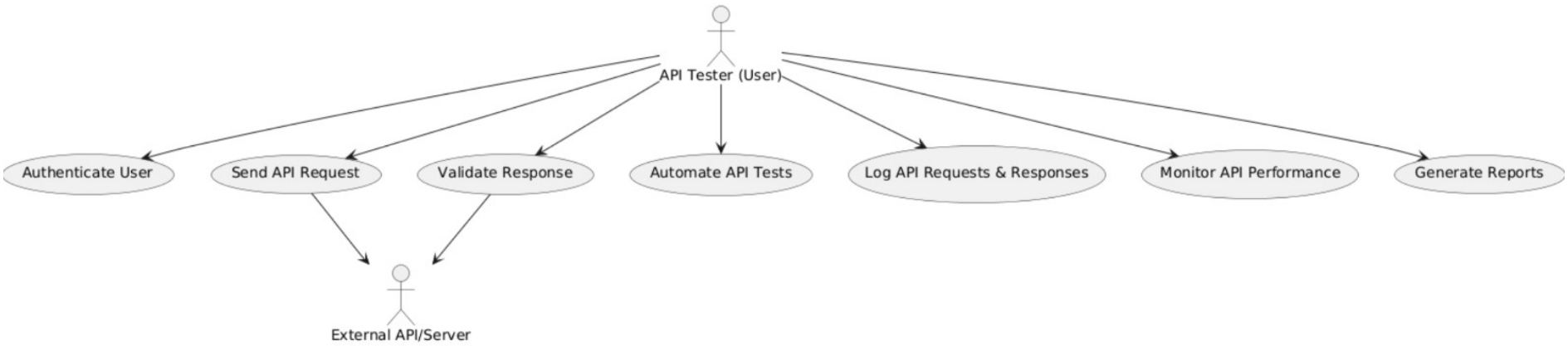
System Architecture



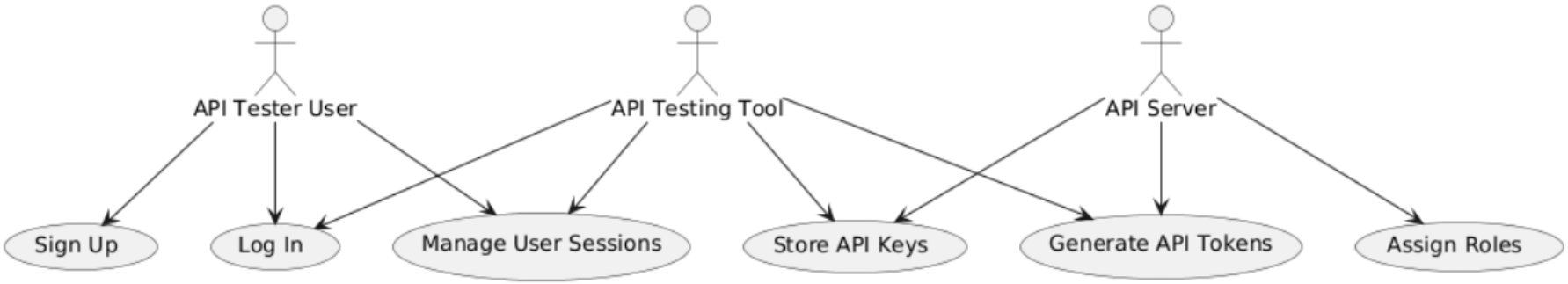
System Architecture

- Client Side: User interacts with the web interface.
- Server Side: Manages API test execution and logging.
- Database Layer: Stores user configurations, test results, and logs.

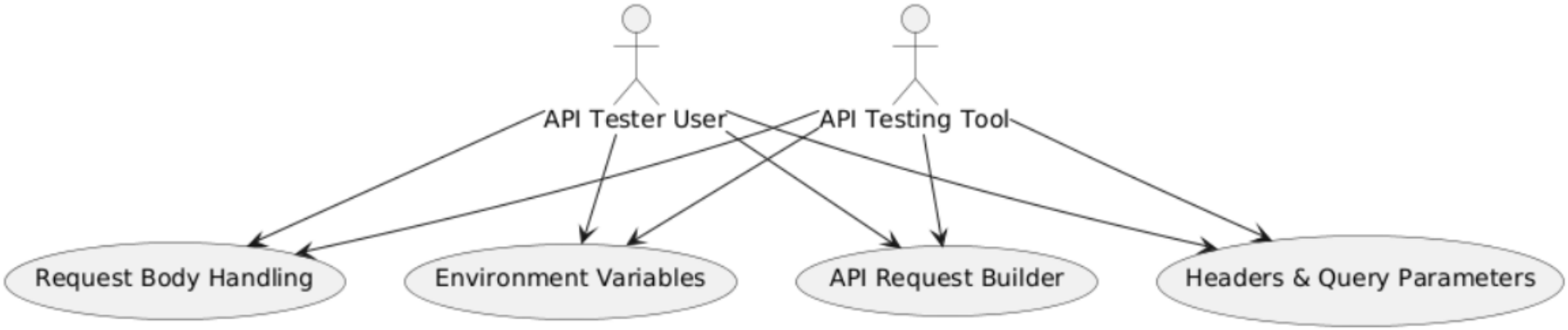
Use Case Diagram / DFD



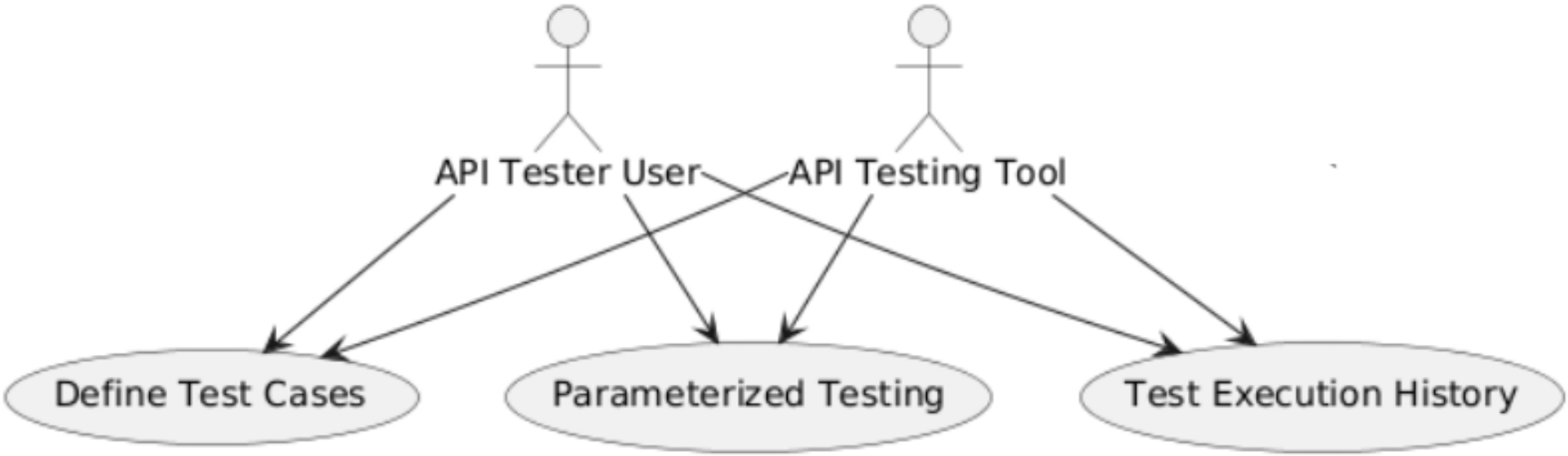
User module


API Tester User
API Testing Tool
API Server
Sign Up
Log In
Manage User Sessions
Store API Keys
Generate API Tokens
Assign Roles

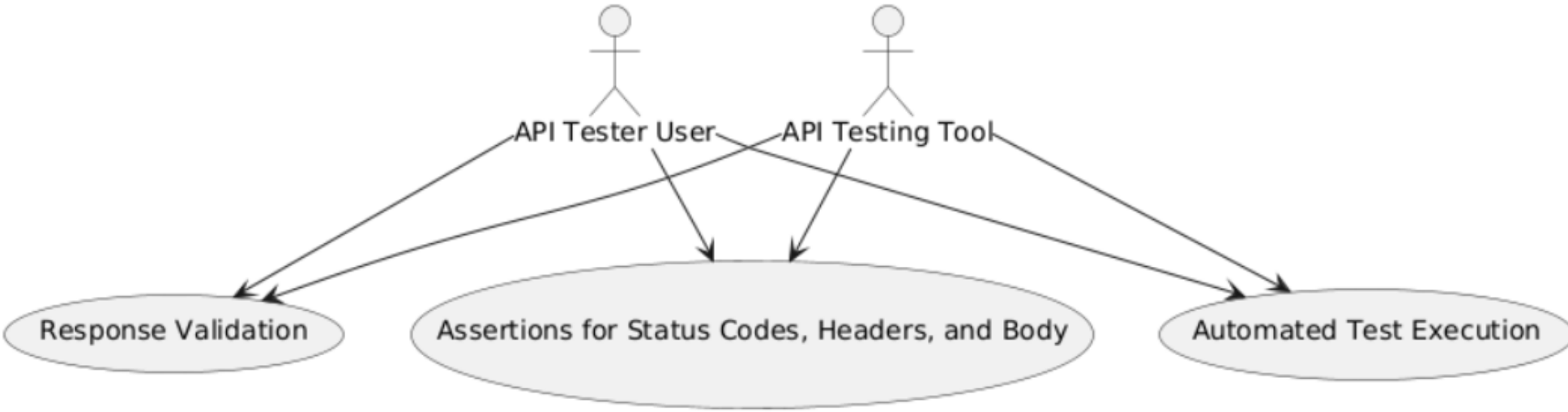
Request module / DFD



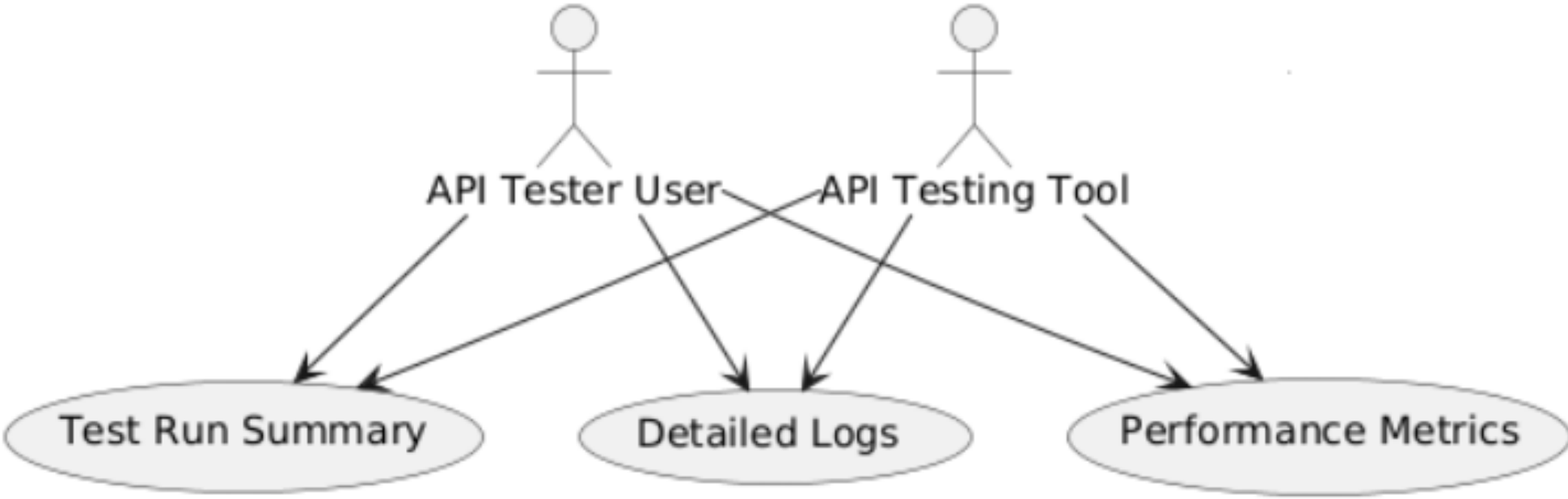
Testcase module



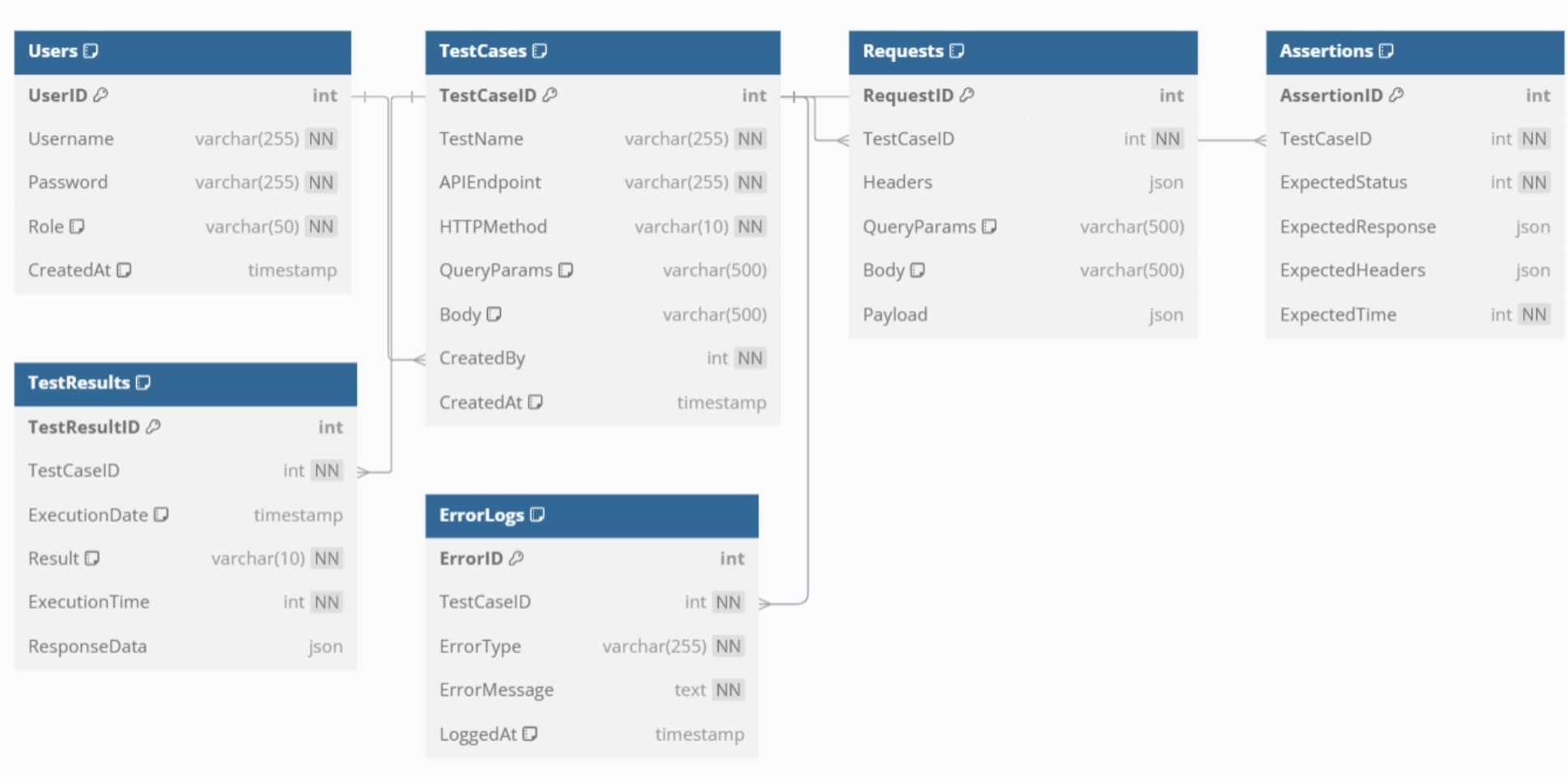
Assertion module



Report module



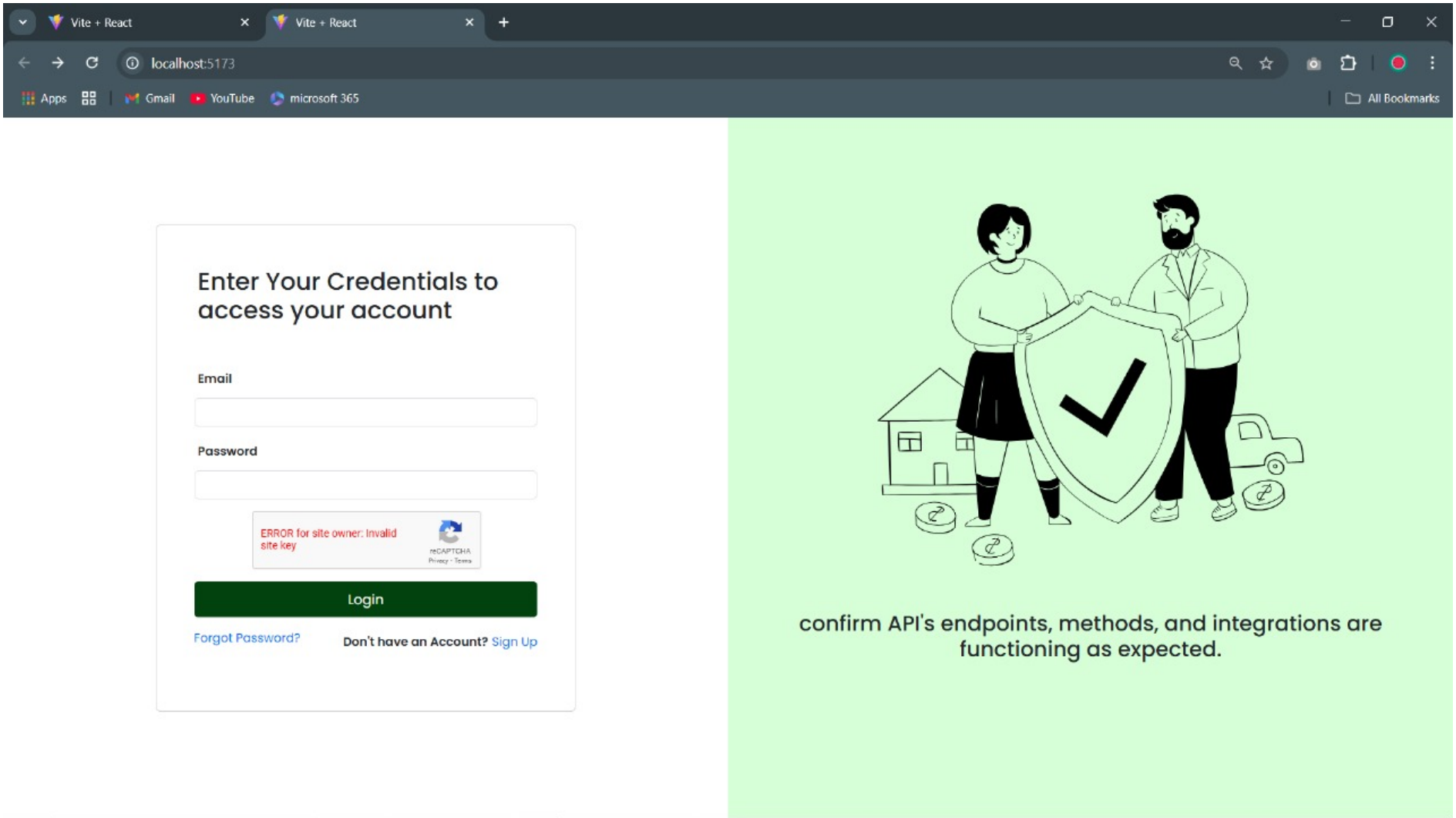
Database Design



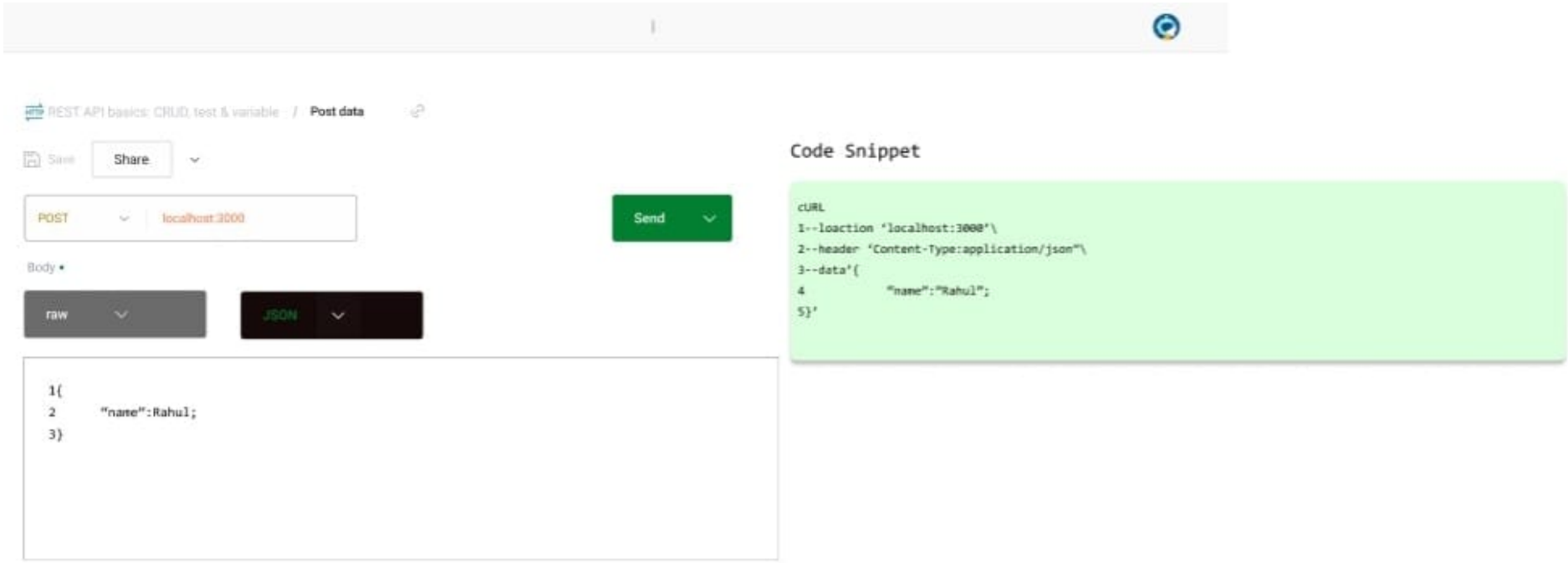
Database Design

- Users Table: Stores user credentials and roles.
- TestCases Table: Contains metadata of test cases.
- Requests Table: Maintains request details.
- Assertions Table: Defines expected response structure.
- Test Results Table: Logs execution results.
- ErrorLogs Table: Tracks errors encountered during testing.

Screenshots



Screenshots



Completion Status

- Requirement Analysis Completed
- System Design Completed
- Backend and Frontend Development In Progress
- Testing and Debugging Ongoing
- Deployment Upcoming

Conclusion

- API Unit Tester automates API testing and reduces manual effort.
- Ensures robust and reliable APIs for seamless application performance.
- Contributes to a smoother and more efficient software development lifecycle.